

Learning and Generalizing Stress Patterns with a Sequence-to-Sequence Neural Network

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Abstract

We present the first application of modern neural networks to the well studied task of learning word stress systems. We tested our adaptation of a sequence-to-sequence network on the Tesar and Smolensky test set of 124 “languages”, showing that it acquires generalizable representations of stress patterns in a very high proportion of runs. We also show that the model, in an agent-based simulation, is biased toward systematic patterns of stress, despite having the expressive power to memorize its training data.

1 Introduction

Some of the earliest work in computational phonology investigated the acquisition and representation of word stress patterns Dresher and Kaye (1990); Gupta and Touretzky (1994). Stress is of interest because the extent of typological variation is relatively well understood, and because learning the patterns is non-trivial in various ways. A considerable amount of more recent work has focused on a data set created by Tesar and Smolensky (2000, henceforth *TS*); see further Jarosz (2013), Jarosz (2015), Boersma and Pater (2016) and Lee et al. (2024). The data set includes 124 languages that can be represented using 12 relatively standard Optimality Theoretic (Prince and Smolensky, 2004) constraints. Past work has tested various algorithms for weighting and ranking constraints to see which performed the best on this dataset (where performance was measured by how many of the 124 languages the models could learn with 100% accuracy).

In this paper, we explore how well a model without constraints, namely a sequence-to-sequence neural network, performs on the 124 languages. Two factors motivate this departure from constraint-based models: (i) a question of whether pre-specified structures like constraints¹ are necessary to represent and learn the

¹For an approach to stress learning that involves constraints that are not pre-specified, see Hayes and Wilson (2008).

stress patterns in the TS data set, and (ii) whether neural networks, which have the expressive power to capture both general and lexically specific patterns will be able to generalize stress patterns to novel data. We find that the sequence-to-sequence net does succeed in learning most of the languages, and that it generalizes to novel data when trained on the 124 TS languages, using an analogue of a training vocabulary and unseen words. We then show, using agent-based modeling, that the network is biased toward more systematic patterns, despite having the expressive power to memorize much or all of its training data.

2 Background

The TS dataset was created to test an approach for handling *hidden structure* in phonology: how does a learner parse a form that it’s being trained on when it hasn’t learned all of the grammatical information needed for parsing in the first place? In the TS languages, this takes the form of stress patterns that are assumed to rely on foot-based structure to place the primary and secondary stress in a word. While the training data for a language includes mappings between underlying forms (strings of light and heavy syllables) to correctly stressed surface forms, that data does not include information about where the feet occur in the correct surface forms. An example of a piece of learning data in one of the TS languages is shown in (1), with *L* representing light syllables and *L1* representing a syllable with primary stress.

$$(1) \quad /L\ L\ L/ \rightarrow [L\ L1\ L]$$

This datum illustrates the ambiguity present when learning stress patterns like these in a foot-based theory—this word could contain a left-aligned foot with iambic stress like $[(L\ L1)\ L]$ or a right-aligned foot with trochaic stress like $[L\ (L1\ L)]$. Each of the 124 languages consists of 62 mappings like this. The 62 input strings are all possible combinations of *Ls* and *Hs* for strings of 2 to 5 syllables in length, plus strings of 6 and 7 *Ls*. For each input string, the candidate set of output strings consists of all possible parsings of the syllables into unary and binary feet, including ones where syllables are left unparsed. There is a minimum of one foot for each word. One of the stresses is designated as primary, and the candidate set has all possible primary stress placements.

Each output string has a corresponding vector of constraint violations, for the 12 constraints shown in (2). Each of the 124 languages in the test set can be generated by some OT ranking of these constraints. That is, some ranking can make a parsed structure optimal that is consistent with the stress pattern in the output of the learning datum, for all 62 target mappings.

- (2) Constraints from the TS Data Set (constraint definitions from Jarosz, 2013)
 - a. *FtBin*: Each foot must be either bimoraic or disyllabic.
 - b. *Parse*: Each syllable must be footed.
 - c. *Iambic*: The final syllable of a foot must be the head.

- d. *FootNon-fin*: A head syllable must not be final in its foot.
- e. *Non-fin*: The final syllable of a word must not be footed.
- f. *WSP*: Each heavy syllable must be stressed.
- g. *WordFoot-R*: Align right edge of the word with a foot.
- h. *WordFoot-L*: Align left edge of the word with a foot.
- i. *Main-R*: Align head foot with right edge of the word.
- j. *Main-L*: Align head foot with left edge of the word.
- k. *AllFeet-R*: Align each foot with right edge of the word.
- l. *AllFeet-L*: Align each foot with left edge of the word.

TS proposed that a learner uses its current grammar to parse a form and then updates its constraint rankings according to that parse. Most subsequent work (with the exception of Jarosz, 2015) has been based on this general premise (Jarosz, 2013; Boersma and Pater, 2016; Lee et al., 2024).

TS found that when they ran their model 10 times on each of the 124 languages in the data set, it achieved perfect accuracy on a language 60.48% of the time. Boersma and Pater (2016) found that when they used a similar parsing strategy, but with numerically weighted constraints instead of ranked ones, and with a stochastic component in the parsing process, languages were learned fully correctly 88.63% of the time. Jarosz (2013) pushed performance on this data set even further, showing that by revising the parsing strategy, success could be achieved 94.19% of the time over 10 runs of the 124 TS languages.

The state-of-the-art on the TS data for constraint-based models (95.73%) was achieved by Jarosz (2015) whose model used a pair-wise ranking grammar with a learning algorithm inspired by expectation maximization (Dempster et al., 1977). This allowed the model to avoid the problem of parsing altogether, since it was able to sample the mappings that various constraint rankings create over the course of acquisition to see which were most likely to improve its performance. Most recently, Lee et al. (2024) showed that a Maximum Entropy learner using expectation maximization achieves a success rate that approaches that of Jarosz (2015): 93% (when weights are initialized at 1).

3 Our Model

While various neural network architectures have been used in phonology, such as feedforward networks (e.g., Gupta and Touretzky, 1994; Moreton, 2012), simple recurrent networks (e.g., Hare, 1990), and convolutional neural networks (e.g., Beguš, 2020), here we focus on the encoder-decoder architecture (Sutskever et al., 2014).

This architecture was originally constructed for machine translation, but is convenient for modeling phonological mappings since it can straightforwardly map between strings of differing lengths, needed for dealing with processes like epenthesis and deletion. This is accomplished by processing the input and output

strings with separate recurrent neural networks. The input is fed into the first network (called the *encoder*), which has no output layer. The recurrent connections of the encoder then pass information about the input to the second network (called the *decoder*), which has no meaningful input, but *does* have an output layer.

A number of studies have shown that when applied to phonological patterns, encoder-decoder networks display similar learning biases to humans (Prickett, 2019, 2021) and also generalize in a human-like way on phonological and morphological tasks (Kirov and Cotterell, 2018; Prickett et al., 2018, see Corkery et al. (2019) and McCurdy et al. (2020) for evidence to the contrary).

In all of the simulations presented here, the network had 2 layers each in its encoder and decoder, with 20 units in each layer, and hyperbolic tangent activation functions throughout. A batch size of 1, was used to minimize the mean squared error between the model’s output and the correct output throughout learning.

4 Methods and results

4.1 Original Tesar and Smolensky (2000) Languages

We first tested our model to see how well it could learn the 124 original languages in the TS data set. In each input string, a timestep for the model represented a single syllable, with a [syllable weight] feature distinguishing between light ($= -1$) and heavy ($= 1$) syllables. In the output, timesteps again represented individual syllables, with the features [stress] and [primary] used to distinguish between syllables with primary stress (values of 1 and 1, respectively), secondary stress (values of 1 and -1 , respectively), and no stress (values of -1 and -1 , respectively).

We ran the model with a learning rate of .0005 for 500 epochs once on each of the 124 languages in the TS set. We found that the network achieved perfect accuracy² in 98.39% (122/124) of the languages. This represents the highest rate of success for any model on this test set. However, it’s unclear whether the network was actually encoding generalizable information, or just memorizing the 62 mappings present in each language, which would be a fairly trivial task for a model with this number of parameters.

Previous research has used the constraints in (2), which are all defined to hold for any string of a particular phonological type. They provide no way of encoding a situation in which two strings of a given type behave differently, as occurs in many real languages (in “exceptions”, or more generally in lexically conditioned patterns). The acquired constraint-based grammars are therefore guaranteed to generalize, though at the cost of not being able to capture lexically conditioned patterns. In what follows, we test whether our learner does learn generalizable representations of the data by including multiple tokens of each type of input string.

²Since the network’s output takes the form of a vector of real-numbered feature values, each mapping was considered correct if every feature in every timestep of its output had the correct sign (positive or negative), given that mapping’s input.

Table 1: Percent of languages with perfect accuracy in training and testing.

Tokens per Type	Training	Testing
3	86.29	44.35
6	98.39	90.32

4.2 Generalization from Tesar and Smolensky (2000) Languages

To test whether the encoder-decoder network was learning generalizable patterns or just memorizing the mappings in each language, we introduced an extra set of “lexical” features to the inputs of the TS data set. These features were implemented as a random, base-2 label for each of the tokens in training, representing the different tokens of each mapping that one would expect in an actual language. For example, the mapping from (1) would have multiple copies in training, each of which had a unique, non-zero label in their input, as illustrated in (3). These are meant to represent multiple words in a language with three light syllables and penultimate stress (like English *banana* and *cabana*).

(3) Examples of Multiple Tokens of the Same Mapping Type

- a. /L L L/₀₁₀₁ → [L **L**1 L]
- b. /L L L/₁₁₁₁ → [L **L**1 L]
- c. /L L L/₀₀₀₁ → [L **L**1 L]

Crucially, each token that belonged to the same mapping type had the same output in all of these simulations (that is, there was no lexical conditioning in any of the stress patterns). We created two sets of data using this system: one that had 3 tokens for each of the 62 mapping types in each of the TS languages and one that had 6 tokens for each of the types.

We ran the encoder-decoder model on these two data sets with a learning rate of .0005 for 200 epochs. At the end of training, we tested the model on 62 novel pieces of data, each of which represented one of the mapping types from training, but with only 0’s for the lexical label features. If the network learned a generalizable pattern from training, it should correctly map all 62 of the novel testing items. The results for training and testing on both data sets are shown in Table 1.

These results suggest that, with enough tokens per type, the network *does* generalize correctly from almost all (112 out of 124) of the languages in the TS data. Additionally, the accuracy on both training and testing increased with the number of tokens per type, suggesting that a number of tokens higher than 6 might allow the model to do even better on both. Natural languages, of course, tend to have more than six words with a given type of stress pattern, at least for shorter words.

4.3 Agent-based Modeling

Unlike many models of phonological learning, neural networks like the one used here can capture patterns that are arbitrarily idiosyncratic. However, many languages

have stress patterns that are systematic and general (as is true for all of the stress patterns discussed in §4.1–4.2). The question remains, then, whether using an encoder-decoder network to model the acquisition of stress patterns leaves this apparent bias for systematicity unexplained.

To test this question, we ran agent-based simulations (Kirby and Hurford, 2002), meant to capture the effect on typology of multiple speakers in a language community interacting with one another (for similar work in the domain of phonology, see Pater and Moreton, 2012; Pater and Staubs, 2015; Hugto, 2020). To do this, we had two encoder-decoder networks (labeled “A” and “B”) each start with a random initialization of weights and alternate being “teacher” and “learner” across trials, with the first teacher generating an output for a randomly produced input. This output then forms the training data for the learner. At the end of 50 iterations, languages generated by the agents are then taken as instances in a typological distribution.

The languages consisted of 100 words that could have initial or final stress, with the encoder-decoder networks taking a word as input and outputting its stress position. Input words were represented as a sequence of phonemes, where a phoneme was a vector of phonological features, with numerical values replacing the [+], and [-] values typically used in phonology (1 and -1 respectively). Phonemes came from the inventory [p, b, t, d, k, g, a, i, e, o, u], all syllables were CV, and word lengths all belonged to the set 2, 4, 6, 8, 10. Each run, a new vocabulary of 100 words was randomly produced by sampling from this space.

We measured the degree of systematicity as the number of words that follow the majority choice (minimum 50, maximum 100). Figure 1 shows the neural networks’ gradual increase over the 50 iterations of teaching and learning toward a more systematic pattern.

The question of why languages tend to be systematic arises for any theory of language that includes an account of lexical or morphological idiosyncrasy. Our agent-based demonstration of an emergent tendency to systematicity is inspired by an earlier similar one in Pater and Moreton (2012), which uses lexically specific constraints as the mechanism for lexical stress in a Maximum Entropy grammar framework. As Pater and Moreton note, there seems to be no empirical work quantifying the degree of systematicity seen cross-linguistically, in stress or in other domains. Linguists seem to have the intuition that languages are regular, and there is empirical evidence of diachronic regularization (Phillips, 1984; Sonderegger and Niyogi, 2013). These agent-based simulations show that adopting a model that allows for idiosyncrasy does not necessarily allow it free rein in subverting systematicity.

5 Discussion

5.1 Comparison with earlier research

Our results on the TS data set with an encoder-decoder model are comparable to the best achieved with constraint based models. It is difficult to compare directly,

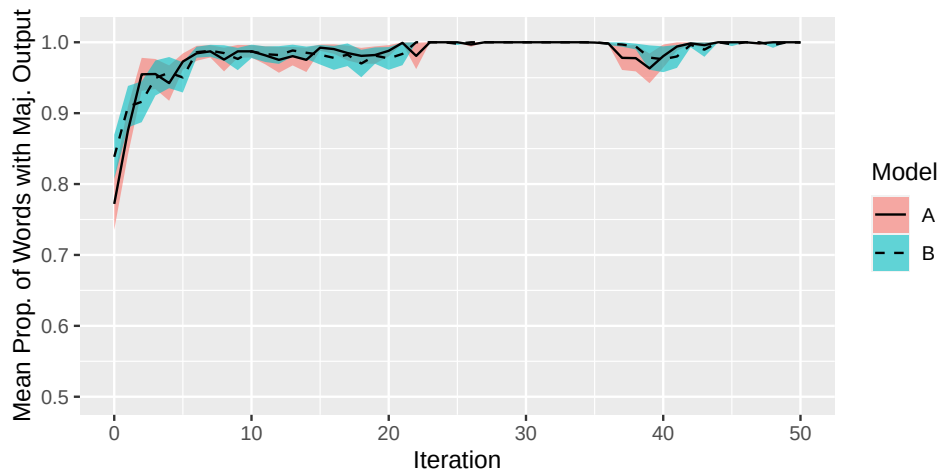


Figure 1: Average proportion of words in each model’s output representing the majority output (either left-aligned or right-aligned stress). Colored regions show 95% confidence intervals.

since the 98.39% accuracy achieved in the first set of simulations, as well as on the training data in the 6 lexical item condition, could be attributable to the model simply representing each of the individual mappings, rather than learning generalizable representations. Nonetheless, the fact that it generated the correct stress pattern for 90.32% of the unseen tokens when there were 6 tokens of each type in the training data shows that it is capable of learning these patterns in a generalizable way with a high degree of accuracy.

None of the prior research on the TS data set provided a means for representing lexical idiosyncrasy, cases where two words of the same syllable shape have different stress patterns. There is a body of prior work on constraint-based approaches to lexical idiosyncratic phonology, however. Tesar (2006) presents an approach to learning exceptions in terms of contrastive specification of underlying features, Pater et al. (2012) propose an alternative that uses constraints on Underlying Representations within a MaxEnt learning framework, Moore-Cantwell and Pater (2016) explore the use of lexically specific constraints in MaxEnt, Hughto et al. (2019) study similar lexically scaled constraints, and Nazarov (2018) presents another approach to learning lexically specific constraints. All of this work has been done on very small systems, and it is not immediately clear how well the proposals will scale up to cases with even the number of constraints in the TS test set, let alone constraint sets that are large enough to deal with the complexity of less idealized individual languages, and of a fuller typology.

5.2 Future Work

A number of avenues exist for future work. The results presented in §4.2 made the TS data set more realistic by introducing multiple lexical tokens for each type of mapping. Making the TS data even more realistic is one potential future direction, for example, by representing inputs and outputs as strings of phonemes rather than just strings of light and heavy syllables.

Another limitation of the TS data set is that it is based on a hypothetical typology based on permuting the rankings of constraints rather than a real-world typology of stress-based patterns. Lee et al. (2024) point out that the languages that a MaxEnt learner tends to fail on in the TS set are likely unattested, raising the possibility that learning explains these typological gaps. Gupta and Touretzky (1994) provide an analysis of their learning results with a neural model that gives an example of how this research could proceed in a neural context.

There is also of course an open question of how substantive tendencies in the typology of attested stress patterns can be explained given a neural model. For example, it is not obvious that it would predict any tendency towards alternating stress patterns. An example of how this could be addressed can be found in Goldsmith (1994).

Finally, computational phonology often involves comparing predictions made by models to human behavior in artificial language learning studies (e.g., Wilson, 2006). Such studies involving stress patterns *do* exist (e.g., Carpenter, 2016), and future work should compare the acquisition and generalization observed in them to that of computational models of stress learning.

5.3 Conclusions

In this paper, we presented results showing that an encoder-decoder neural network can successfully learn a variety of stress patterns. Using the Tesar and Smolensky (2000) data set (a commonly cited benchmark for models of stress), we were able to show that the network outperformed past models when tested on how many of the languages in the data set it could acquire perfectly.

We then created an extension of this data set that included multiple tokens of each relevant mapping type in the 124 languages, and differentiated these tokens using lexically specific labels for each word. When the model was given data that included six tokens for each mapping type from the original data set, its performance on novel test items was comparable to past, state-of-the-art approaches.

Finally, agent-based modelling showed that, despite having the expressive power to capture idiosyncratic stress assignment, when two networks interacted with one another (alternating teacher/student roles), the networks shifted together toward more systematic patterns of stress.

Taken together, these results show that (i) pre-specified constraints are not necessary for a model to successfully learn and generalize stress-based patterns and (ii) while the neural network we used had the ability to simply memorize

the mappings we were training it on, it generalized to novel data, and displays a tendency towards systematicity in an agent-based setting.

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